

Bringing the Heat: day-ahead demand forecasts for heat-pump homes

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Task: forecast how much electricity 375 heat-pump households will draw from the grid on the next day, so E.ON can buy the right amount on the day-ahead market. **Result:** one gradient-boosting model for all households predicts each household’s daily demand with a **mean absolute error of 5.8 kWh** (average demand 26.9 kWh per day) and $R^2 = 0.82$. For all households combined, the daily error is 7.8 % ($R^2 = 0.94$).

Daily demand of all households, test year (Mar 2023 - Feb 2024)



Data and setup

- **Data:** 15-min grid consumption of 375 households with > 1 year of data and survey metadata (118 with PV, 99 without, 158 unknown), 8 hourly weather stations.
- **Cleaning:** columns read by name (their order differs between files); missing and impossible readings (> 40 kWh) dropped, never imputed; weather timestamps mark the end of the hour and are shifted; sunshine for the 3 stations without a sensor is filled from the other stations and flagged.
- **Day-ahead rule:** the forecast for day D only uses data up to the end of day D-1 (yesterday’s load and weather, the load one week earlier). Nothing from day D.
- **Split by time:** train before 1 March 2023, test on the following 12 months (to evaluate effectiveness for entire year). Hyperparameters tuned on Jan-Feb 2023, never on the test year.

Model (Level 0)

One **global gradient-boosting model** GBM for all households with 26 features: load of the same 15 min yesterday and one week earlier, yesterday’s weather, time of day, week-day, month, PV flag, heat-pump optimisation visit and building survey answers. Most important (permutation importance): yesterday’s and last week’s load, yesterday’s mean temperature, living area, time of day. A hyperparameter search on Jan-Feb 2023 kept the default settings.

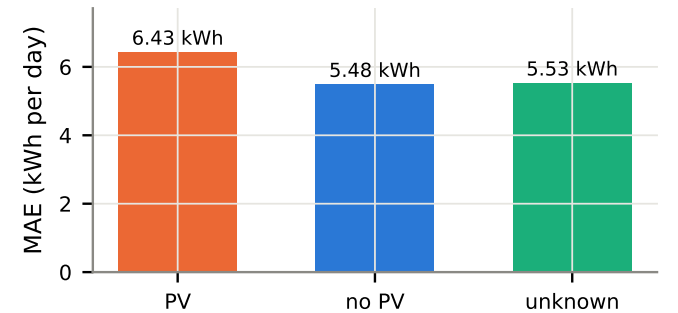
Results: daily demand per household, test year

Model	MAE (kWh per day)	R^2
Gradient boosting	5.79	0.819
Baseline: same day last week	7.87	0.631

The baseline repeats each household’s demand from the same day one week earlier. Gradient boosting **cuts the daily error by 26 %** ($7.87 \rightarrow 5.79$ kWh) and raises R^2 from 0.63 to 0.82, because it learns how demand reacts to temperature, time of day and household characteristics. Without a weather forecast for day D, it still reacts to weather changes about a day late (see the cold snaps in the chart above). For all households combined, the errors largely cancel: the daily error is 750 kWh on an average of 9.6 MWh (7.8 %).

PV vs no PV (Level 1)

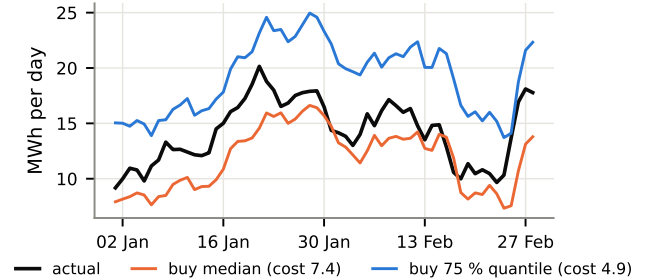
Daily error per household by PV status



PV households have a $\approx 17\%$ higher daily MAE (6.43 vs 5.48 kWh): their grid draw depends on the next day’s sunshine, which the model only knows from yesterday. Unknown-status households behave like no-PV ones.

Uncertainty and buying decision (Levels 2–3)

Buying the median vs the 75 % quantile (Jan-Feb 2023)



Quantile gradient boosting: 80 % intervals contain 78 % of the readings (Jan-Feb 2023 validation). If buying too little costs 3× more than buying too much, the optimal amount is the 75 % quantile ($3/(3+1)$). Buying it instead of the median **cuts the daily cost by 34 %**; summed household medians fall short on 56 of 59 days.

Limitations and next steps

- **No weather forecasts in the data:** with day D’s measured weather as a perfect forecast, the daily error dropped from 11.3 % to 4.2 % in an earlier experiment. A real weather forecast is the biggest lever.
- Next: quantiles of the *total* demand (summed household quantiles over-buy), PV-specific models, real market prices.